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Write your **student number** in the boxes above. **Letter**

Food Studies

Question and Answer Book

VCE Examination – Monday 17 November 2025

- Reading time is **15 minutes**: 9.00 am to 9.15 am
- Writing time is **1 hour 30 minutes**: 9.15 am to 10.45 am

Materials supplied

- Question and Answer Book of 24 pages
- Multiple-Choice Answer Sheet

Instructions

- Follow the instructions on your Multiple-Choice Answer Sheet.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

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Section A (20 questions, 20 marks) 2–7
Section B (10 questions, 70 marks) 8–22

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VICTORIAN CURRICULUM AND ASSESSMENT AUTHORITY

VICTORIA State Government

Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** or that **best answers** the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.

Question 1

In a study of 158 parents interviewed in Australian supermarkets, 73% of parents reported their children making food requests during shopping. Of these requests, 88% were for discretionary foods, such as chocolates and lollies.

Source: Adapted from C Collins, 'The power of pestering', World Cancer Research Fund, <www.wcrf.org>

Which of the following is the most likely reason for children making food requests?

A. The children are experiencing hunger when in the supermarket.

B. The children are influenced by their sensory appreciation of food.

C. Chocolate and lollies are every child's favourite foods.

D. The children are experiencing a lack of satiety when in the supermarket.

Question 2

Which statement best describes an individual's psychological response when following a restrictive diet?

A. An individual may develop an intense focus on food.

B. An individual may become less socially engaged with others.

C. An individual may get upset when they follow a restrictive diet and stop losing weight.

D. An individual may feel disappointed when they break a restrictive diet.

Do not write in this area.

Question 3

> The November 2024 Inquiry into Food Security in Victoria found that many stakeholders advocated for a place-based approach to food literacy. A place-based approach to food literacy focuses on developing food knowledge and skills within the context of a specific community, utilising local resources, relationships and cultural practices.

Source: Adapted from Legislative Council Legal and Social Issues Committee, *Food security in Victoria*, November 2024, p. 90, <www.parliament.vic.gov.au>

Which one of the following is an example of a place-based approach to food literacy designed to improve food security?

A. a nutritional information handout from a national supermarket about a new food

B. a program that encourages and educates people in the use of locally grown produce

C. a food safety poster designed for school canteens

D. a television cooking show that focuses on the use of Australian native ingredients

Question 4

According to Food Standards Australia New Zealand (FSANZ), which one of the following statements is an example of a general health claim on a food label?

A. Calcium is important for strong bones and teeth.

B. Calcium reduces the risk of osteoporosis.

C. Diets low in sodium help lower blood pressure.

D. High sodium intake causes hypertension.

Question 5

The *Australian Dietary Guidelines* and the 'Australian Guide to Healthy Eating' (both part of the 'Eat for Health' program) are credible sources of information because they

A. were published in 2013 by Nutrition Australia.

B. promote good nutrition and health.

C. contain reviews of information from nutrition experts and food-based social media influencers.

D. contain evidence from randomised controlled trials and systematic literature reviews.

Question 6

A new diet trend makes the claim 'Eating raw foods leads to better health and weight loss'.

This claim can be evaluated by checking

A. which raw food items are allowed in the diet and whether the diet incorporates a range of vegetables.

B. scientific studies related to the diet to establish whether the diet is supported by credible evidence.

C. whether the diet is based on research conducted by professionals and whether it is recommended by them.

D. how widespread the diet is and whether it is endorsed by well-known people in the media.

Do not write in this area.

Question 7

Which of the following statements best explains how the current food system may contribute to overconsumption of food?

A. Foods are rejected at the farm when they could be sold at the supermarket at a cheaper price.

B. Transportation costs for discretionary foods are cheaper than for fresh foods.

C. Discretionary foods are often positioned prominently at the end of supermarket aisles and at the checkout.

D. Discounted food can be over-purchased and then wasted at home.

Question 8

The nutrition information shown in the table below is for a Greek yoghurt product.

NUTRITION INFORMATION

Servings per package: approx. 5 Serving size: 170 g

	Avg. quantity per serving	Avg. quantity per 100 g
Energy	400 kJ (96 cal)	235 kJ (56 cal)
Protein	15.9 g	9.4 g
Fat, total	LESS THAN 1 g	LESS THAN 1 g
- saturated	LESS THAN 1 g	LESS THAN 1 g
Carbohydrate	6.8 g	4.0 g
- sugars	6.1 g	3.6 g
Dietary fibre, total	LESS THAN 1 g	LESS THAN 1 g
Sodium	70 mg	41 mg
Calcium	204 mg (26% RDI▲)	120 mg

▲% Recommended dietary intake per serving

Based on the nutrition information panel above, what is a possible nutrition content claim that could be accurately featured on this food product?

A. The product is a good source of fibre.

B. The product is high in calcium.

C. Diets high in calcium may reduce the risk of osteoporosis.

D. Calcium is good for bones and teeth.

Do not write in this area.

Question 9

bananas

sweet potatoes

garlic

Sources (from left): Maks Peoplenko/Shutterstock.com; mahirart/Shutterstock.com; Denis180787/Shutterstock.com

Fruits and vegetables such as the three foods shown in the images above

A. are prebiotic foods that feed beneficial bacteria in the gut and contribute to optimal health.

B. contain the same amount of fibre, which starves the beneficial bacteria in the gut.

C. are probiotic foods that enhance beneficial gut bacteria and have significant health benefits.

D. are highly processed foods that have a detrimental impact on our physical and mental health.

Question 10

A new bread brand includes wholegrains as well as resistant starch, which functions as a prebiotic that improves digestive health.

Resistant starch improves digestive health because it

A. is fermented by bacteria in the large intestine.

B. provides beneficial bacteria that our bodies absorb.

C. promotes satiety.

D. nourishes the cells that line the small intestine.

Question 11

The chemical breakdown of protein into amino acids in the stomach involves

A. protease and amylase.

B. chyme and peristalsis.

C. amylase and lipase.

D. hydrochloric acid and pepsin.

Question 12

Which of the following statements best describes the physiological response in individuals with gluten intolerance?

A. An allergic reaction triggers anaphylaxis.

B. A digestive reaction causes inflammation of the gastrointestinal tract.

C. An immune reaction causes damage to the gastrointestinal tract.

D. An enzymatic reaction results in cramps and diarrhoea.

Do not write in this area.

Question 13

Fermentable oligosaccharides, disaccharides, monosaccharides and polyols (FODMAP) intolerances are a result of

A. a genetic predisposition that inhibits the consumption of carbohydrates.

B. poor absorption of specific carbohydrates in the small intestine.

C. an inability to break down specific carbohydrates due to insufficient lactase.

D. a condition of the small intestine where villi become inflamed and damaged.

Question 14

The Australian Research Council (ARC) Industrial Transformation Training Centre for Uniquely Australian Foods works with Aboriginal and Torres Strait Islander communities to research and grow native foods in a way that respects the communities' knowledge, rights and culture. One project is developing a low-sugar Kakadu plum soft drink to improve health and create jobs for Aboriginal and Torres Strait Islander people.

Source: Adapted from Y Sultanbawa, 'Bush foods bring health, cultural and economic benefits for Indigenous Australians', Australian Research Council, <www.arc.gov.au>

How might the Kakadu plum soft drink project support the social role of food in Aboriginal and Torres Strait Islander communities?

A. by reducing sugar consumption among the wider population and promoting better health for all

B. by prioritising economic benefits for Aboriginal and Torres Strait Islander communities through food production and job creation

C. by fostering cultural pride within Aboriginal and Torres Strait Islander communities through sharing traditional knowledge

D. by strengthening cultural identity and community connections through using traditional native ingredients

Question 15

An example of a sociocultural concern that can influence individual food choice is

A. environmental sustainability.

B. religious beliefs.

C. consumer welfare.

D. technological advances.

Question 16

Which one of the following types of information about the effectiveness of a nutrient supplement could be considered valid?

A. the labelling of the nutrient supplement's list of ingredients and how they improve health outcomes

B. the nutrient supplement's website, which provides multiple statements from users of the nutrient supplement about the benefits they have experienced using the product

C. an article in a blog that is written by a dietitian and peer-reviewed, summarising research completed on the nutrient supplement's effectiveness

D. a recommendation from a gym instructor

Do not write in this area.

Question 17

One solution for overcoming the loss of biodiversity in Australia is

A. combining Aboriginal and Torres Strait Islander and Western knowledges.

B. culling kangaroos to reduce their impact on the land.

C. collecting data on climatic changes.

D. reducing quarantine controls on imports.

Question 18

According to the Australian Good Meat website, it currently takes less water for cattle to grow 1 kg of beef than it did five years ago.

One possible reason for the reduction in the use of water in beef production in Australia is

A. outbreaks of disease and biosecurity concerns.

B. the use of genetics and breeding to improve the suitability of cattle to their environment.

C. the promotion of irrigation for pasture production.

D. increased demand for plant-based meat alternatives.

Use the following information to answer Questions 19 and 20.

The Internet of Things (IoT) refers to a network of physical objects, such as food products and packaging, which are embedded with sensors and software. These objects collect and exchange data over the internet, enabling real-time monitoring, automation and improved decision-making throughout the food system, from production and storage to distribution and consumption.

Question 19

Being able to track the location of food products in real time is likely to be an effective way of helping to feed the growing world population because

A. consumers can track when their food will arrive in the supermarket.

B. tracking produce in real time enables farmers to see where their products have been sent.

C. real-time tracking helps identify problems in a supply chain that may lead to food loss.

D. reducing food losses caused by items going missing or being stolen may help lower the cost of food.

Question 20

Which of the following best demonstrates an ethical contribution of the IoT?

A. Food production is increased and jobs are lost due to mechanisation.

B. Food is delivered efficiently to vulnerable communities, with minimal food loss.

C. The IoT is used to track consumer behaviour.

D. The data provided on food storage is ignored to reduce transportation costs.

End of Section A

Do not write in this area.

Section B

Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.

Question 1 (6 marks)

In the table below, explain one role that each of the accessory organs has in the digestion of fat.

Accessory organ	Role of each organ in the digestion of fat
liver	
gall bladder	
pancreas	

Do not write in this area.

Question 2 (10 marks)

A university student lives in a four-bedroom shared housing arrangement at the university, which includes a shared kitchen and storage space. The students all share a single microwave oven and a refrigerator. As a third-year full-time student who works part-time at a fast-food restaurant, the student is very concerned about their own eating habits. They often rely on the convenience of fast food from their workplace and ready-to-eat meals from the supermarket for their daily meals.

a. Describe **two** examples of how living within a shared house could influence the student's food accessibility. 4 marks

Question 2 continues on the next page.

Do not write in this area.

b. Explain why the use of a timetable to allocate time for all students to use the kitchen space could influence healthy eating. 3 marks

c. Analyse **one** way the social factor of education could influence the food choices of the students. 3 marks

Do not write in this area.

Question 3 (4 marks)

Yuzu is a citrus fruit known for its strong, tangy lemon-like flavour. Yuzu is not commonly eaten whole due to its sourness, but is widely used in juices, sauces and desserts. An online publication claims yuzu is a 'superfood' that helps reduce inflammation, promotes heart health, and is high in antioxidants and vitamins. The term 'superfood' is not a scientific classification and is used as a marketing term.

Sources: Adapted from T La Forge, '33 top food trends for 2025', Cozymeal, <www.cozymeal.com/magazine/top-food-trends>; E Mandl, '13 emerging benefits and uses of yuzu fruit', Healthline, <www.healthline.com/nutrition/yuzu-fruit>

Assess the information about yuzu using the criteria of language use and source.

Do not write in this area.

Question 4 (8 marks)

The following table lists the number of serves of grain (cereal) foods – preferably wholegrain – required by adults in various life stages.

Age/Stage	Number of serves
19–50 years	6
51–70 years	4
70+ years	3
pregnancy	8.5

Source: Adapted from 'Recommended number of serves for adults', *Australian Dietary Guidelines*, <www.eatforhealth.gov.au>

a. Describe **one** reason why a pregnant woman needs more serves of grain (cereal) foods than a 19- to 50-year-old woman. 2 marks

b. Using the nutritional rationale of the 'Australian Guide to Healthy Eating', discuss the ways in which grain (cereal) foods can optimise health. 4 marks

Do not write in this area.

c. Outline **one** way an individual can manage their allergy to wheat. 2 marks

Question 5 (4 marks)

Discuss **two** reasons why consumers may choose to purchase foods from farms and primary producers that follow ethical practices, even though prices may be higher.

Do not write in this area.

Question 6 (8 marks)

Competitive eating is a sport that challenges contestants to compete against each other by eating large quantities of highly processed food, in a short period of time, for prize money.

a. Common foods consumed during eating competitions include hot dogs and doughnuts.

Discuss how the regular consumption of hot dogs and doughnuts in eating competitions may influence an individual's mental health. 4 marks

Do not write in this area.

b. Participating in competitive eating competitions may contribute to the overconsumption of food.

Discuss the relationship between the overconsumption of food and its effect on physical health. 4 marks

Do not write in this area.

Question 7 (8 marks)

How many vegans are there in Australia?

According to research, 2.5 million Australians now have diets in which the food is all, or almost all, vegetarian or vegan[1] – that's 12.1% of adults in Australia. In the last decade, the number of people following a plant-based diet has increased by about 50%.

Source: Adapted from J van der Linden, 'Veganism in Australia: The rise of plant-based eating', Veg Me Up, <www.vegmeup.com.au>

[1]**vegan** – a person who does not eat meat, fish, dairy or any animal products

a. Identify and describe two examples of how the media influences Australians to adopt a vegetarian or vegan diet. 4 marks

Example 1 _____

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Example 2 _____

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Do not write in this area.

b. Discuss **two** ways in which the growing trend of vegetarian and vegan diets may have an effect on an individual's health or wellbeing. 4 marks

Do not write in this area.

Question 8 (8 marks)

A 2024 study found that approximately one in 10 adolescents has used weight-loss products and nutrient supplements. Protein powders and other nutrient supplements such as creatine also have wide appeal. Creatine is a supplement used to improve strength, muscle mass and overall performance during exercise. Creatine is also found naturally in foods such as red meat and fish.

Research shows that girls are more at risk of choosing to take weight-loss supplements than boys. However, a growing problem among boys is the use of protein powders and creatine products, which are compounds that supply energy to muscles.

Source: Adapted from E Hemendinger and K Suleta, 'Dietary supplements and protein powders fall under a "wild west" of products that necessitate caveats and caution', *The Conversation*, <https://theconversation.com>

a. Apply the criteria of ethics and effectiveness of the product to claims made by weight-loss and nutrient supplement companies targeting adolescents. 4 marks

Ethics _____

_____

_____

_____

_____

_____

Effectiveness of the product _____

_____

_____

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Do not write in this area.

b. Evaluate the role of commercial gain when assessing the validity of claims made by weight-loss and nutrient supplement companies. 4 marks

Question 9 (4 marks)

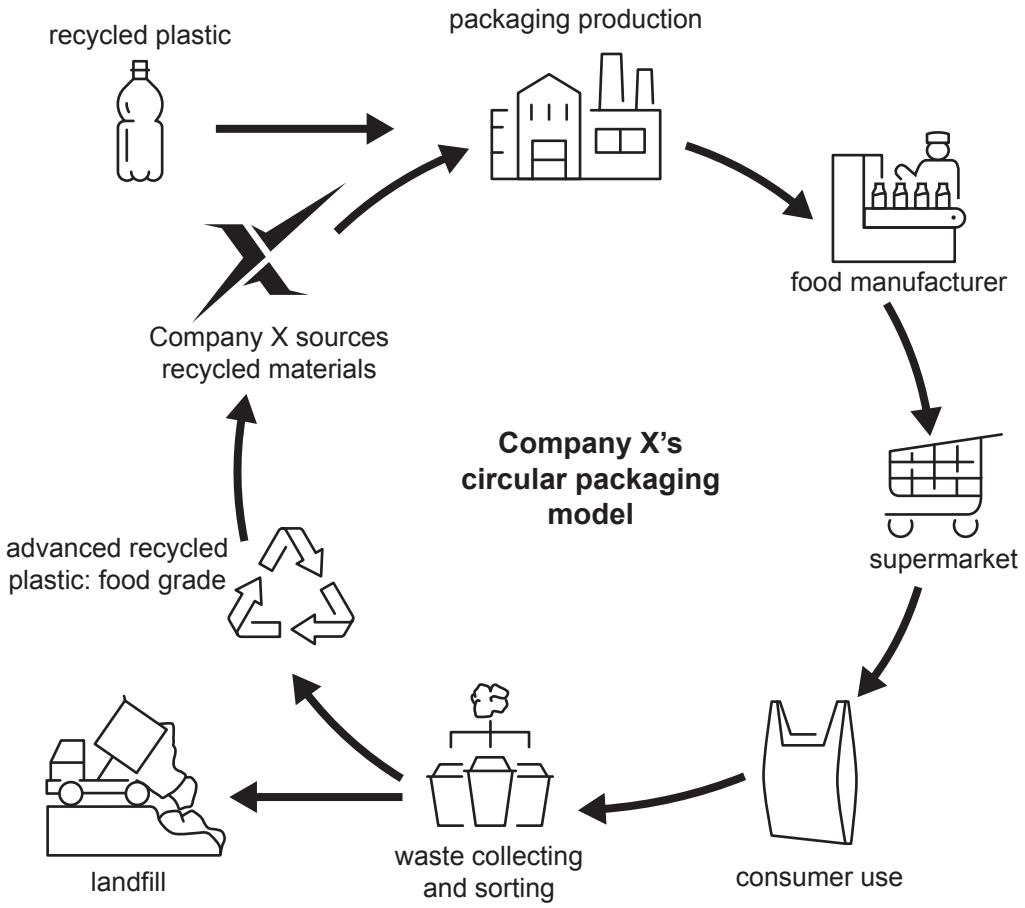
Consumer activism is a type of political influence on our food system.

a. Explain **one** action consumer activists may take to raise public awareness of excessive consumption of sugar from sweetened drinks. 2 marks

b. Describe **one** possible impact on the food system as a result of increased public awareness of excessive consumption of sugar from sweetened drinks. 2 marks

Do not write in this area.

Question 10 (10 marks)



Company X produces recycled plastic packaging for a frozen vegetable line

In 2024, Company X committed to sustainable packaging by incorporating recycled plastic materials, starting with microwavable plastic steam bags. This year, two products within Company X's microwavable frozen vegetables range will contain 22% recycled content in their packaging. The company has plans for further expansion of recycled packaging content across all their products. Company X is a multinational company that manufactures and distributes worldwide.

The new plastic packaging will feature a QR code linking to a video about the brand's sustainability efforts. This initiative allows consumers to learn more about Company X's commitment to the circular economy. A circular economy is a system where products are reused, repaired or recycled instead of being thrown away, to reduce waste and save resources. Circular packaging follows this idea by using packaging that can be recycled or reused multiple times, rather than ending up in landfill after a single use.

Company X highlighted that there are challenges in sourcing recycled plastic for frozen products; however, it aims to increase its recycled plastic content by 2026, as supply improves.

Company X noted that incorporating 22% recycled plastic into its packaging will eliminate 4.6 tonnes of virgin plastic annually. This follows an earlier initiative by the company to switch to a single type of recyclable plastic for packaging in 2021.

Sources: Adapted from *Packaging News*, <www.packagingnews.com.au>; (images clockwise from top left) Nadiinko/Shutterstock.com; Skylines/Shutterstock.com; matsabe/Shutterstock.com; Skylines/Shutterstock.com;Nadiinko/Shutterstock.com; Skylines/Shutterstock.com; Skylines/Shutterstock.com; Nadiinko/Shutterstock.com; Shahanawaz Bhuyan/Shutterstock.com

Do not write in this area.

Evaluate Company X's pathway to improving environmental and economic sustainability in food manufacturing. Consider how Company X supports food citizenship and food sovereignty.

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Question 10 continues on the next page.

End of examination questions

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